

Depth-Assisted Semi-Supervised RGB-D Rail Surface Defect Inspection

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Abstract—Visual-based methods for rail surface defect inspection (RSDI) effectively improve the limitations of manual inspection, as they can intuitively display the locations and segmented areas of sensitive defects. The RGB-D RSDI task, which leverages the complementarity between RGB and depth (D) image information to enhance detection performance, has attracted widespread attention and achieved significant development. However, existing methods primarily depend on fully supervised training strategies that necessitate a substantial number of manually annotated pixel-level labels to supervise model training. Undoubtedly, extensive manual annotation is exceedingly time-consuming and labor-intensive, particularly considering the irregular shapes and textures of surface defects on rails, further compounding the burden of manual labeling. Therefore, in this paper, we aim to introduce the semi-supervised learning paradigm into this task. Towards the semi-supervised RGB-D RSDI task, a specific semi-supervised network for this task and an effective cross-modal fusion module are crucial to ensuring detection performance under the constraints of limited labeled samples. Thus, we propose a Depth-assisted Semi-Supervised RGB-D RSDI network (DSSNet) to simultaneously alleviate the annotation burden and achieve satisfactory detection performance. Specifically, adhering to the consistency training paradigm, we construct a semi-supervised RGB-D RSDI architecture for this task by optimizing structures, perturbation mechanisms, loss settings, etc. Furthermore, we propose a Depth-assisted Multi-scale Cross-modal Fusion Module (DMCFM) that conducts multi-scale exploration and cross-modal complementarity fusion with the assistance of depth. Comprehensive experiments demonstrate that, compared to the latest 14 state-of-the-art fully supervised methods, the proposed DSSNet achieves highly competitive results while effectively alleviating an 80% annotation burden.

Index Terms—Rail surface defect inspection, RGB-D images, semi-supervised learning, salient object detection.

I. INTRODUCTION

RAIL transportation has become one of the primary modes for both travel and the transportation of goods

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in contemporary times. Consequently, the task of rail surface defect inspection (RSDI) has naturally garnered significant attention, and with the rapid advancement of deep learning, the vision-based rail surface defect inspection task has made substantial progress [1], [2], [3], [4].

Due to the diversity of factors causing rail surface defects, the final defects exhibit significant variations in shape and area. Meanwhile, the rail surface foreground and background share substantial similarities, significantly increasing the difficulty of surface defect inspection. For instance, traditional manual inspection methods [5] are inefficient and insensitive, while some nondestructive measurement methods [6] (ultrasonic testing, acoustic emission testing, and magnetic flux leakage) face limitations in intuitively presenting defect shapes. Consequently, more advanced vision-based algorithms [7], [8] have emerged successively. These algorithms primarily leverage rail surface images (RGB and Depth) to achieve more effective and sensitive defect localization and segmentation.

In recent years, there has been a continuous emergence of RGB-based algorithms for rail surface defect detection [7], [9], [10], [11], [12], [13], [14]. Zhang et al. [15] introduced a novel RSDI scheme for limited samples, treating defective images as sequential data and classifying pixel lines. However, the specificity of the task results in acquired RGB images of the rail surface being extremely different (e.g., in color, texture, and contrast) from those in a natural scene. This leads to a weak differentiation of the acquired RGB images in the foreground (defective areas) from the background (non-defective areas). Therefore, achieving reliable detection performance based on RGB images alone for the task of RSDI is challenging.

With the rapid development of the sensor industry, the effective supplementation of depth (D) information, which enhances the advantages of capturing scene 3D structure, has been introduced into the task, promoting the flourishing development of RGB-Depth RSDI tasks [16], [17], [18], [19], [20]. However, existing RGB-D RSDI methods [2], [3], [16], [17], [18], [19], [20] primarily rely on fully supervised training strategies to achieve satisfactory detection performance. Less satisfying is the fact that vision-based RSDI tasks still present the following two challenges: 1) The training strategy of fully supervised methods heavily relies on pixel-level annotations from a substantial amount of finely labeled data. The complexity of working conditions makes collecting defect data challenging, and the randomness and variability of defect shapes make manual labeling extremely time-consuming and labor-intensive. 2) The extreme similarity between the foreground and background on the rail surface

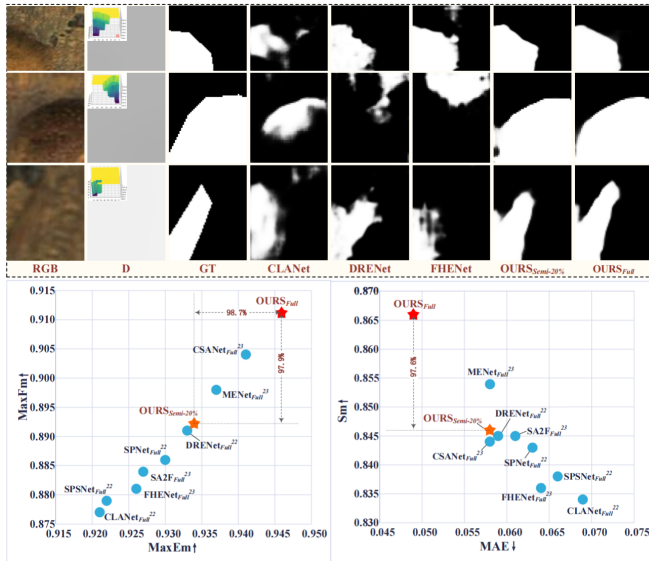


Fig. 1. Qualitative and quantitative comparisons with the latest fully supervised methods. The proposed DSSNet, under the premise of alleviating **80%** of the annotation burden, achieves performance gains of over **97%** compared to our state-of-the-art fully supervised version, while also delivering highly competitive results compared to the latest fully supervised methods.

in terms of color, texture, and contrast will interfere with the effective detection of RGB images. While depth cameras can effectively complement the 3D structure of the scene, the minimal relative distance between the foreground and background in this specific task will also amplify the impact on the contribution of depth. Therefore, with respect to the RGB-D RSDI task, there is an urgent need to explore how to achieve satisfactory detection performance while minimizing the annotation burden as much as possible.

Based on the considerations mentioned above, in this paper, we attempt to introduce the semi-supervised learning paradigm into this task, forming a semi-supervised RGB-D RSDI, with the goal of alleviating the annotation burden. In the context of semi-supervised RGB-D RSDI, a specific semi-supervised network architecture and an effective cross-modal fusion module that integrates complementary RGB and depth information are crucial for obtaining satisfactory detection performance with limited annotated samples. Therefore, we first introduce the consistency training paradigm [21] as the foundation for the semi-supervised network. Subsequently, we further optimize the architecture, perturbation mechanisms, and loss settings for the specific task of RSDI, constructing an effective semi-supervised bimodal RGB-D RSDI network.

Considering the subtle differences in the vertical direction between the foreground and background on the rail surface, depth images struggle to dominate both in visual representation and in making practical contributions to detection performance (As shown in Fig. 1). This challenge is widely acknowledged in related RGB-D salient object detection (SOD) tasks [22], [23], [24], [25], [26], even though in such tasks, there is a clear depth gap between the foreground and background. However, existing methods often treat depth and RGB images equally for exploration and fusion [2], [16], [18], [19], [20]. Therefore, more effective exploration and fusion of complementary information from both modalities,

further promoting satisfactory detection performance under limited sample constraints, remains a critical challenge for semi-supervised RGB-D RSDI tasks. To address this challenge, we propose a Depth-Assisted Multi-Scale Cross-Modal Fusion Module (DMCFM). This module conducts multi-scale feature exploration and complementary fusion for both modalities through depth assistance and RGB guidance. Simultaneously, DMCFM adopts a holistic multi-scale exploration pattern, effectively improving the local connectivity limitation introduced by convolution operations in cross-modal fusion.

Ultimately, by embedding DMCFM into the constructed semi-supervised RGB-D RSDI network architecture, we have established the Depth-Assisted Semi-Supervised RGB-D RSDI Network (DSSNet). As shown in Fig. 1, the proposed DSSNet achieves a dual success of effectively alleviating annotation burden and obtaining satisfactory detection performance. Our contributions are summed up as follows:

- 1) To the best of our knowledge, this is the first attempt to introduce the semi-supervised learning paradigm to alleviate the heavy annotation burden in the RGB-D bimodal RSDI task.
- 2) By optimizing the architecture, perturbation mechanisms, and loss settings, we have constructed an effective semi-supervised architecture specifically tailored for the RGB-D RSDI task.
- 3) We propose DMCFM by analyzing the relative limitations of the two modalities, effectively performing multi-scale feature exploration and complementary fusion, while alleviating the local connectivity introduced by convolution operations.
- 4) Extensive experiments on a unified benchmark dataset have validated the effectiveness of our approach. Compared to **14** fully supervised methods, the proposed DSSNet achieves highly competitive results while effectively alleviating an **80%** annotation burden. Compared to our state-of-the-art fully supervised version, alleviating 80% of the annotation burden has yielded over **97%** performance gains across multiple metrics.

II. RELATED WORK

A. RGB-D Rail Surface Defect Inspection

The RGB-D Rail Surface Defect Inspection (RGB-D RSDI) task, which leverages the complementarity between depth and RGB image information to facilitate rail surface defect detection, has gained widespread attention in recent years. A large-scale RGB-D RSDI dataset was introduced by [16], fostering the emergence of numerous promising algorithms for this task [2], [3], [16], [17], [18], [19], [20]. Yang et al. [18] developed a Contour and Semantic Feature Alignment Fusion Network (CSANet), utilizing bidirectional feature alignment to explore the internal consistency of cross-modal features from both contour and semantic perspectives. In consideration of the speed of operation and computational resource issues for deployment on mobile devices, Zhou and Hong [27] proposed a lightweight feature hierarchical exploration network (FHENet) for real-time RSDI. Yan et al. [20] proposed a Specificity Autocorrelation Integration Network (SAINet)

for RSDI, enhancing defect detection performance by exploring autocorrelation features within a single modality and the specificity of each modality. However, existing methods are all based on fully supervised training strategies, requiring a substantial amount of manually annotated pixel-level labels to supervise model training. Meanwhile, current approaches treat both modalities equally during cross-modal fusion, making it challenging to effectively leverage the relative differences in information contributions between modalities for exploring and fusing cross-modal.

B. Rail Surface Defect Inspection

A significant number of algorithms have emerged from research on RSDI relying on visual RGB images. Nieniewski [9] presented a system for RSDI and shape extraction utilizing morphological operations. Piao et al. [10] proposed a horizontal U-shaped magnets-based motion-induced eddy current array (MIECA) method to detect rail surface defects with the HSHS capability. Ni et al. [12] achieved effective partitioning of edge features to accomplish detection of railroad surface defects by effectively extracting edge features and establishing a uniform background on the railroad surface to eliminate the effects of uneven illumination. Zhang et al. [28] proposed a multi-model RSDI system based on CNNs (MRSDI-CNN) from the standpoint of studying the squat on the rail surface. Meanwhile, semi-supervised learning has also been applied in single-modal defect segmentation and anomaly detection tasks. Wu et al. [29] proposed a simple and efficient semi-supervised defect segmentation method, named Perturbed Progressive Learning. It ensures performance by decoupling the predictions of student and teacher networks and encouraging consistency across various perturbations in a broader stage-wise scope. Ren et al. [30] proposed a pavement anomaly detection network (PADNet), which is a semi-supervised learning approach based on generative adversarial networks for identifying pixel-level anomalous image segments. These methods primarily focus on single-modal RGB data and utilize knowledge distillation or generative adversarial networks to construct semi-supervised architectures.

C. RGB-D Salient Object Detection

Salient object detection aims at detecting and segmenting the most attractive objects or regions in an image or video, which is of great interest for the task of rail surface defect inspection. Meanwhile, multi-modal saliency detection tasks have been extensively studied [21], [22], [23], [24], [25], [26], [27], [31], [32], [33], [34], [35], [36], [37], [38]. Zhou et al. [39] combined a gated attention mechanism and a linear fusion method to construct a dual-stream interactive recursive feature-reshaping network (IRFR-Net). Wu et al. [40] observed that the depth information of color images can enhanced the feature representation associated with SOD, and thus proposed an implicit depth restoration (IDR) technique to enhance the feature representation of RGB-D SOD in mobile networks. There have been some promising works have started to explore weakly-supervised [41], [42], [43], [44] and semi-supervised learning [45], [46] architecture

to alleviate the annotation burden. Li et al. [43] proposed a weakly-supervised RGB-D salient object detection model based on scribble supervision, achieving effective multi-modal representation learning through the regularization of inter-modal mutual information. Zhu et al. [46] introduced weak supervision signals into a basic SSLA for learning. However, existing methods [45], [46] rely on obtaining a large amount of unlabeled samples in addition to fully supervised labeled training data to achieve superior detection performance. They have not effectively addressed the goal of alleviating the burden of annotation.

III. METHODOLOGY

A. Overview

1) *Motivation*: The importance of the task of RSDI is self-evident in ensuring the safety of railway transportation. However, less satisfying is the fact that vision-based RSDI tasks still present the following two challenges: 1) The training strategy of fully supervised methods heavily relies on pixel-level annotations from a substantial amount of finely labeled data. The complexity of working conditions makes collecting defect data challenging, and the randomness and variability of defect shapes make manual labeling extremely time-consuming and labor-intensive. 2) The extreme similarity between the foreground and background on the rail surface in terms of color, texture, and contrast will interfere with the effective detection of RGB images. While depth cameras can effectively complement the 3D structure of the scene, the minimal relative distance between the foreground and background in this specific task will also amplify the impact on the contribution of depth.

Therefore, the challenge lies in how to effectively alleviate the annotation burden without incurring substantial performance losses and how to leverage limited annotated samples for effective cross-modal information exploration and fusion, ultimately promoting the achievement of satisfactory detection performance. Given the aforementioned considerations, we initially constructed a semi-supervised architecture for the RGB-D RSDI task to effectively alleviate the burden of annotation while attaining foundational performance. Subsequently, we introduced DMCFM to further conduct multi-scale information exploration and complementary fusion based on the relative contributions of the two modalities, thereby achieving satisfactory performance with limited annotated samples.

2) *Architecture*: The network architecture of the proposed DSSNet is illustrated in detail as in Fig. 2. From the perspective of data samples, we randomly select 1/5 of the data from the existing and widely used RGB-D rail surface defect inspection training set as labeled samples (i.e., using labeled GT) and the remaining as unlabeled samples (i.e., not using labeled GT). All training data pass through the encoder, which first performs feature extraction for both modalities. This is followed by cross-modal and cross-level fusion of the extracted features. We employ the ConvNeXt architecture [47], a family of pure ConvNet models, as the backbone of the network to extract multilevel features separately for the two modalities (R_i and D_i , ($i = 0, \dots, 3$)). Our proposed DMCFM

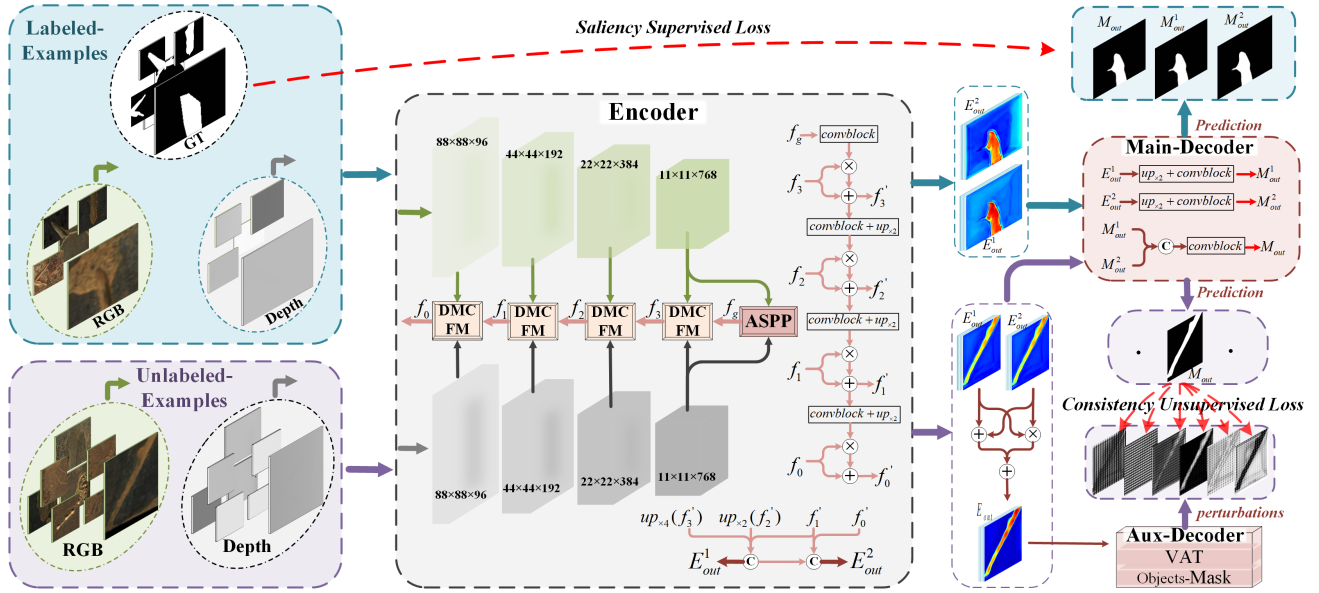


Fig. 2. The overall architectural pipeline of the proposed depth-assisted semi-supervised rgb-d rail surface defect inspection network (DSSNet).

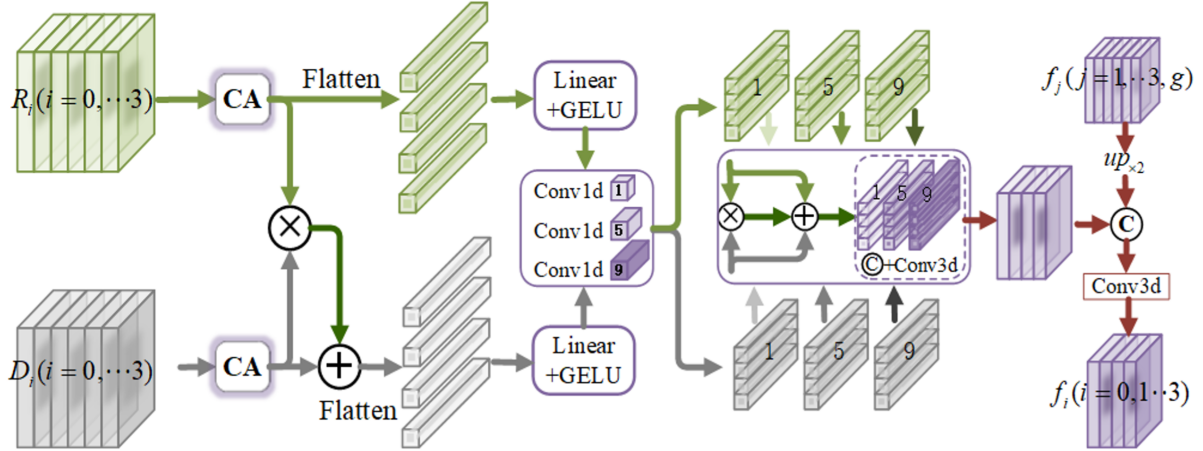


Fig. 3. The detailed processing pipeline of the proposed depth-assisted multi-scale cross-modal fusion module (DMCFM).

integrates the multilevel features of the two modalities and performs a simple cross-level fusion to obtain the two output features of the encoder after the integration. For the labeled samples, they undergo decoding and prediction by the main decoder after encoding, and the saliency loss between the predicted results and the labels is calculated. For the unlabeled samples, the prediction is generated by the main decoder and the perturbation is added by the auxiliary decoder, whereas the consistency loss between the two outputs is calculated. The entire network is trained through joint learning, allowing continuous utilization of unlabeled samples to enhance learning ability of the model and achieve superior defect detection performance under the limitation of labeled samples.

B. Depth-Assisted Multi-Scale Cross-Modality Fusion Module

It is evident that the captured rail surface images differ significantly from natural scene images; for instance, the RGB images of the rails lack distinct color,

texture, contrast, and other salient features. Additionally, considering the narrow spatial distance between the foreground and background of the rail surface, it becomes crucial to use depth information as a complementary modality for better localization, thereby assisting the RGB modality in completing the defect inspection. In light of the above discussion, we propose a depth-assisted multiscale cross-modal fusion module. As shown in Fig. 3, this module conducts multiscale feature exploration and cross-modal integration using RGB guidance and depth assistance. Initially, a commonly used channel attention module is employed to attention-weight the feature channels of both modalities. Subsequently, cross-modal features are integrated through element-level multiplication with depth, presented in the form of residuals. This achieves RGB primary guidance along with depth-assisted cross-modal fusion. The expression can be represented as:

$$\begin{aligned} x_r &= CA(R_i), (i = 0, \dots, 3), \\ x_d &= (x_r \otimes CA(D_i)) \oplus CA(D_i), \end{aligned} \quad (1)$$

where $CA(\cdot)$ represents channel attention module, $\oplus$ and $\otimes$ denote element-level summation and multiplication, respectively. For all multimodal tasks, achieving multiscale mining of proprietary information between modalities and effective fusion of cross-modal features is critical.

Initially, we employ a multilayer perceptron to comprehensively establish information correlations between features. Subsequently, we explore multiscale information by applying multi-kernel 1D convolution to the constructed holistic features. This approach ensures the consistent information of overall feature correlation among the features, facilitating the extraction of multi-scale effective information.

$$x_m^k = \text{conv}_1^k(\text{mlp}(x_m)), (m = r, d | k = 1, 5, 9), \quad (2)$$

where conv_j^k represents that the convolution has dimension j and the convolution kernel is k . where $\text{mlp}(\cdot)$ represents the commonly used multilayer perceptron. After the establishment of the overall correlation of features between two modalities and the exploration of multiple scales, we immediately implemented the fusion of cross-modal features at the same scale as well as across scales. An efficient cross-modal feature fusion of two modalities has been achieved so far, which is activated in the form of RGB guidance as well as depth assistance, continued with feature holistic correlation mining and multi-scale exploration, and finally completed with cross-modal and cross-scale feature fusion.

$$f^{cmf} = \text{conv}_3^3([\text{cat}(x_r^k \otimes x_d^k) \oplus x_r^k \oplus x_d^k]), (k = 1, 5, 9), \quad (3)$$

where $[\cdot]$ represents the concatenation of channel dimensions.

While deeper level features contain more obvious semantic information, shallower level features are rich in detail information. Mining and fusing high-level semantic information first and then supplementing the detail information by continuous upsampling has been considered as a more reasonable and effective way for defect inspection. For better inheriting the high-level semantic information from the fused deep-level features and more effectively supplementing the detailed information of the current-level fused features, we up-sample the high-level semantic information and then fuse the cross-level features by channel concatenation with convolution and strictly limit the number of channels of the final output features. The detailed operation can be expressed as follows:

$$f_i = \text{conv}_3^3([\text{up}_{\times 2}(f_{i+1}), f^{cmf}]), (i = 0, \dots, 3). \quad (4)$$

where $\text{up}_{\times 2}(\cdot)$ represents upsampling with a factor of 2.

We further visualize the feature maps of the input and output of the DMCFM. As shown in Fig. 4, depth images play an assisting role in the fusion process. Moreover, this visualization further validates that our proposed DMCFM effectively explores and fuses cross-modal information through depth assistance and semantic-guided decoding.

C. Main-Decoder and Aux-Decoder

Boosting the learning ability of the network with the help of massive unlabeled samples is the dominant idea of semi-supervised learning and the original intention of the proposed approach. As shown in Fig. 2, our encoder achieves

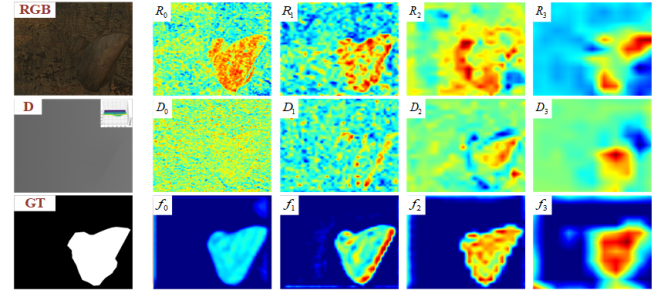


Fig. 4. Visualization of the input and output heatmaps of the DMCFM.

bottom-up feature extraction as well as top-down cross-modal and cross-level fusion. Specifically, the proposed DMCFM achieves effective cross-modal fusion, and simple cross-level fusion is used to integrate the cross-modal fused features to obtain the output of the encoder. The cross-level fusion process can be represented as:

$$f'_3 = f_3 \oplus f_3 \otimes CB(f_g),$$

$$f'_i = f_i \oplus f_i \otimes CB(\text{up}_{\times 2}(f'_{i+1})), (i = 0, 1, 2), \quad (5)$$

$$E_{out}^1 = [\text{up}_{\times 4}(f'_3), \text{up}_{\times 2}(f'_2), f'_1],$$

$$E_{out}^2 = [E_{out}^1, f'_1, f'_0]. \quad (6)$$

where $CB(\cdot)$ represents the convolutional block, which is composed of a convolution, a batch normalization, and a RELU activation function.

For labeled samples, the proposed main decoder is mainly responsible for interpolating the fused features from the encoder to recover the resolution and generate multiple predictions.

$$M_{out}^1 = CB(\text{up}_{\times 4}(E_{out}^1)),$$

$$M_{out}^2 = CB(\text{up}_{\times 4}(E_{out}^2)),$$

$$M_{out} = CB([M_{out}^1, M_{out}^2]). \quad (7)$$

The hybrid loss is used to measure the proximity between the prediction maps and the labels, and the hybrid loss is a combination of the commonly used Binary Cross-Entropy (BCE) loss and the IOU loss. The labeled samples produce losses at multiple outputs of the main encoder noted as l_1 , l_2 and l_0 , respectively. For the main prediction output M_{out} will simultaneously use boundary loss to augment the constraints on the prediction boundary l_b [48], and the overall supervised loss is expressed as:

$$l_{sup} = l_0 + \alpha l_1 + \beta l_2 + \lambda l_b. \quad (8)$$

With the final optimization of the network, α and β are set to 0.8, λ is set to 0.5.

For unlabeled samples, predictions are generated by the main decoder (M_{out}^{un}) and perturbations are added by the auxiliary decoder, respectively, ultimately by constraining the consistency of the two outputs. The output of the encoder is integrated before being fed into the auxiliary decoder.

$$E_{out}^{un} = (E_{out}^1 \oplus E_{out}^2) \oplus (E_{out}^1 \otimes E_{out}^2). \quad (9)$$

The auxiliary decoder consists mainly of the object mask [49] and the VAT [50]. Given the importance of

contextual relationships for scene understanding will result in a network that may be overly dependent on these correlations. Some work [21] has introduced object masks to limit these to mask foreground objects. VAT is introduced to push the output to be isotropically smoothed around each data point. The normalized DICE loss [51] is utilized to constrain the consistency loss of the two auxiliary decoders with the output of the main decoder after adding perturbations, denoted as l_m and l_v , respectively.

$$l_{un\ sup} = \gamma l_m + \varphi l_v. \quad (10)$$

With the final optimization of the network, γ and φ are set to 3. We employ joint learning to train both labeled and unlabeled samples, and the total training loss can be expressed as:

$$l = l_{sup} + \omega l_{un\ sup}. \quad (11)$$

where ω is set to 5 and it is warmed up by 30 epochs.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

1) *Implementation Details*: The proposed DSSNet is based on the Pytorch framework while using a pre-trained model of ConvNeXt on the ImageNet dataset, and all experiments are done uniformly on a single NVIDIA 3090 GPU. The batchsize is set to 8 during training, and 170 epochs are trained at the same time. The Adam optimizer is used for optimizing our networks, and the initial learning rate is initially set to 1e-4, the momentum is 0.9, and the learning rate is guaranteed to decay both modalities to 352×352 and use data enhancement such as random flip, random rotation and random crop.

2) *Datasets*: For fairness and intuitiveness of the comparison, we maintain consistency with the training and testing datasets used by the extant RGB-D RSDI method, again using the NEU RSDDS-AUG dataset [16] (1500 pairs as the training set and 362 pairs as the testing set). Specifically, we randomly selected 1/5 in the training set as the labeled sample set (i.e., 300 pairs of RGB and D images along with their corresponding labels) and the remaining in the training set as the unlabeled sample set (i.e., 1200 pairs of RGB and D images without using their corresponding labels.).

3) *Evaluation Metrics*: The comparison of quantitative metrics is a key criterion to measure network performance. In order to demonstrate more sufficiently and comprehensively the advantages of the proposed approach, we use 10 evaluation metrics mentioned in the field. PR curve, the newly proposed E-measure ($AdpE_m$, $MeanE_m$ and $MaxE_m$) [52], S-measure (S_m) [53] score and weighted F-measure (W_F_m) [54], Mean Absolute Error (MAE) [55], F-measure ($AdpF_m$, $MeanF_m$ and $MaxF_m$) score [56]. The lower the value of MAE and the higher the value of other indicators, the better the performance.

The PR curve is to binarize the salient map under different probability thresholds (0 ~ 1) and compare it with the ground truth map.

$$MAE = \frac{1}{w \times h} \sum_{x=1}^w \sum_{y=1}^h |S(x, y) - Y(x, y)|. \quad (12)$$

where w and h are the width and height of the image.

a) *E-measure* (E_m): The enhanced alignment method considers both local pixel values and image-level mean values which include three main commonly used metrics $AdpE_m$, $MeanE_m$ and $MaxE_m$,

$$E_m = \frac{1}{w * h} \sum_{i=1}^w \sum_{j=1}^h \gamma_{FM}(i, j). \quad (13)$$

where γ_{FM} denotes the enhanced-alignment matrix.

b) *F-measure* (F_m): It is a comprehensive indicator that considers accuracy and recall which include three main commonly used metrics $AdpF_m$, $MeanF_m$ and $MaxF_m$.

$$F_m = \frac{(1 + \beta^2) \cdot precision \cdot recall}{\beta^2 \cdot precision + recall}. \quad (14)$$

where β^2 is set to 0.3.

c) *W_F* (W_F_m): It extends the basic quantities to non-binary values and determines the weight error according to their location and neighborhood.

$$W_F = \frac{(1 + \beta^2) \cdot precision^\omega \cdot recall^\omega}{\beta^2 \cdot precision^\omega + recall^\omega}. \quad (15)$$

where β^2 is set to 1.

d) *S-measure* (S_m): The spatial structure similarity is evaluated by combining the structural similarity of regional perception and the structural similarity of object perception.

$$S_m = \alpha * S_o + (1 - \alpha) * S_r. \quad (16)$$

where α is set to 0.5.

B. Comparison of Method Performance

Due to the absence of relevant weakly supervised or semi-supervised methods for addressing the RGB-D RSDI task, we conducted all comparative experiments against more challenging fully supervised methods. Table I clearly shows the quantitative metrics of the proposed DSSNet compared with the latest 14 relevant fully supervised methods. Specifically, CLANet [16], DRENet [17], FHENet [27], CSANet [18], MENet [2] and SA2F [3] are the latest proposed fully-supervised methods for RGB-D RSDI, and the remaining methods DMRA [57], HAINet [58], S2MA [59], CONet [60], BBSNet [61], SPSNet [62], SPNet [63] and CATNet [64], are the fully supervised RGB-D bi-modal salient object detection methods that are extremely relevant to this study.

1) *Quantitative Comparison*: Quantitative comparisons with 15 relevant fully supervised methods, as shown in Table I, reveal that our proposed semi-supervised version of DSSNet still achieves highly competitive results while effectively alleviating 80% of the annotation burden. Moreover, our fully supervised version of DSSNet, using the same amount of training data as the fully supervised methods, demonstrates a significant performance advantage. Concretely, comparing 14 related fully supervised methods, the proposed semi-supervised version has a slight performance advantage in 9 quantitative metrics, demonstrating the effectiveness of our proposed deep-assisted semi-supervised network architecture. The advantage is even more pronounced when the network

TABLE I

QUANTITATIVE COMPARISON RESULTS OF DIFFERENT MODEL METHODS; RED AND BLUE INDICATE THE TOP TWO INDICATORS IN ORDER.
 FULL: FULLY SUPERVISED (TRAINING USING 1500 PAIRS OF RGB AND D IMAGES ALONG WITH THEIR CORRESPONDING LABELS),
 SEMI: SEMI-SUPERVISED (TRAINING USING 300 PAIRS OF LABELED DATA AND 1200 PAIRS OF UNLABELED DATA)

Methods	RGB-D SOD								RGB-D RSDI						OURS _I Semi ^{20%}	OURS _{II} Full ^{100%}
	DMRA	HAINet	S2MA	CONet	BBSNet	SPSNet	SPNet	CATNet	CLANet	DRENet	FHENet	CSANet	MENet	SA2F		
	Full ₁₉	Full ₂₁	Full ₂₀	Full ₂₀	Full ₂₁	Full ₂₂	Full ₂₂	Full ₂₃	Full ₂₂	Full ₂₂	Full ₂₃	Full ₂₃	Full ₂₃	Full ₂₃		
AdpEm $\uparrow$	0.823	0.714	0.752	0.862	0.839	0.855	0.867	0.868	0.863	0.868	0.866	—	—	—	0.871	0.872
Sm $\uparrow$	0.735	0.717	0.775	0.786	0.827	0.838	0.843	0.861	0.834	0.845	0.836	0.844	0.854	0.845	0.846	0.866
W_Fm $\uparrow$	0.713	0.741	0.690	0.800	0.841	0.855	0.868	0.866	0.845	0.867	0.856	—	—	—	0.871	0.892
AdpFm $\uparrow$	0.764	0.790	0.689	0.830	0.861	0.804	0.878	0.885	0.862	0.879	0.870	—	—	—	0.882	0.900
meanEm $\uparrow$	0.815	0.781	0.789	0.865	0.904	0.916	0.924	0.929	0.912	0.929	0.921	—	—	0.923	0.931	0.942
MaxEm $\uparrow$	0.834	0.828	0.863	0.879	0.909	0.922	0.930	0.942	0.921	0.933	0.926	0.941	0.937	0.927	0.934	0.946
MeanFm $\uparrow$	0.760	0.758	0.743	0.829	0.856	0.864	0.877	0.883	0.863	0.878	0.869	—	—	0.870	0.879	0.897
MaxFm $\uparrow$	0.783	0.805	0.817	0.834	0.867	0.879	0.886	0.903	0.877	0.891	0.881	0.904	0.898	0.884	0.892	0.911
MAE $\downarrow$	0.140	0.171	0.141	0.101	0.073	0.066	0.063	0.055	0.069	0.059	0.064	0.058	0.058	0.061	0.058	0.049

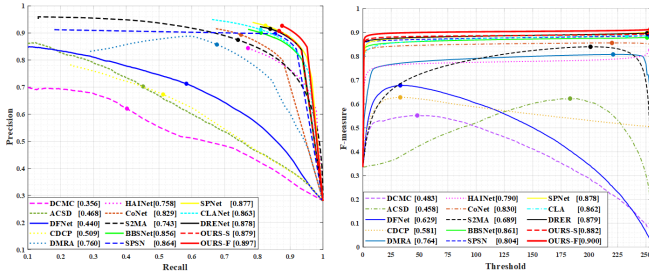


Fig. 5. Quantitative comparison results between our proposed method and the latest methods. The first column is Precision (vertical axis) Recall (horizontal axis) curves, and the second column shows the F-measure scores of the deep learning-based methods under different thresholds.

is trained using the fully supervised approach: Comparing the suboptimal methods, there are 1.2% and 1.3% improvement in S_m and $MeanE_m$, respectively. On some relatively more dominant indicators, there are 2.4%, 2.1% and 1.9% improvement in W_F_m , $AdpF_m$ and $MeanF_m$, respectively. Fig. 5 shows the PR curves and threshold-FM curves for all compared methods, and it can be seen that the proposed DSSNet achieves a parcel of the existing compared methods. In conclusion, our approach achieves more advantageous defect segmentation and inspection performance, proving the effectiveness of our proposed semi-supervised architecture and the main module of the design.

2) *Qualitative Comparison*: Fig. 6 shows the comparison of the defect inspection prediction maps of the proposed method with 13 relevant fully supervised methods. The comparative prediction maps show that our method has a more significant segmentation advantage, and further validating the effectiveness of our network architecture as well as the main modules.

Both the qualitative and quantitative comparisons further validate the effectiveness of our approach. Compared to 14 fully supervised methods, the proposed DSSNet achieves highly competitive results while effectively alleviating an 80% annotation burden. Compared to our state-of-the-art fully supervised version, alleviating 80% of the annotation burden has yielded over 97% performance gains across multiple metrics.

C. Ablation Analysis

Table II further conducts experiments to elucidate the contributions of the proposed key modules. All ablation experiments use the same training set and nine evaluation metrics to assess performance. At the same time, Fig. 7 further illustrates the visualizations of samples when the main modules are ablated.

1) *Effectiveness of the Semi-Supervised Architecture*: Comparing S1 and S2, as well as S3 and S4 in Table II, it can be observed that under the same annotated samples, our semi-supervised approach introduces additional unlabeled samples (the remaining 80% samples) without increasing the annotation burden. This auxiliary training leads to a significant performance improvement for the model. It effectively demonstrates the effectiveness of our semi-supervised architecture designed for RSDI. The sample visualizations in Fig. 7 further validate that the semi-supervised approach, formed by using additional unlabeled samples, further promotes effective segmentation. We further compared with semantic segmentation methods that are also based on consistency training. As shown in Table III, our method exhibits a significant advantage across various metrics. For example, it achieved a 4% performance advantage on most metrics. This further indicates the effectiveness of our approach for the RSDI task.

2) *Effectiveness of the DMCFM*: Comparing S1 and S3, as well as S2 and S4 in Table II, it can be observed that under the same data training strategy, whether it is a semi-supervised or fully supervised mode, the introduction of the proposed DMCFM significantly benefits the model's performance. For example, the quantitative comparison between S2 and S4 shows that the introduction of DMCFM can bring over 3% performance gains across most metrics.

3) *Analysis of the Proportion of Labeled Samples*: We conducted experiments on the proportion of labeled samples in the semi-supervised training mode to further validate the effectiveness of our semi-supervised approach. As shown in Table IV, we achieve basic detection performance when alleviating 90% of the annotation burden, and highly competitive results compared to existing fully supervised methods when alleviating 80%. Moreover, the gains obtained by our model gradually increase with the proportion of labeled samples.

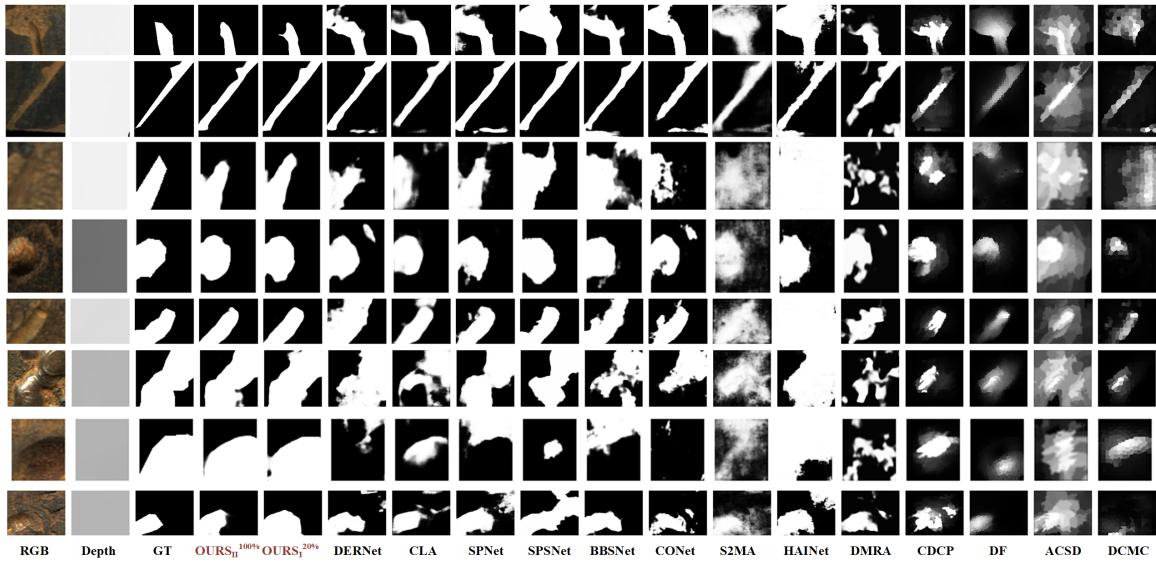


Fig. 6. Visual comparisons between the proposed semi-supervised and fully supervised versions of DSSNet and 13 relevant fully supervised methods.

TABLE II
COMPARISON OF THE DIFFERENT CONTRIBUTION METRICS OF THE MAIN MODULES

Ablation-Settings	Baseline	DMCFM	NEU RSDDS-AUG								MAE↓
			AdpEm↑	Sm↑	W_Fm↑	AdpFm↑	MeanEm↑	MAxEm↑	MeanFm↑	MaxFm↑	
S1-Full ^{20%}	✓		0.820	0.812	0.745	0.847	0.887	0.907	0.842	0.870	0.092
S2-Semi ^{20%}	✓		0.833	0.817	0.819	0.855	0.894	0.914	0.851	0.866	0.083
S2+ASPP	✓		0.837	0.822	0.825	0.858	0.898	0.910	0.850	0.868	0.080
S3-Full ^{20%}	✓	✓	0.851	0.821	0.779	0.861	0.897	0.913	0.854	0.864	0.081
S4-Semi ^{20%}	✓	✓	0.871	0.846	0.871	0.882	0.931	0.934	0.879	0.892	0.058
S5-RGB=Depth	✓	✓	0.864	0.838	0.862	0.873	0.923	0.927	0.871	0.883	0.064
S6-Full ^{100%}	✓	✓	0.872	0.866	0.892	0.900	0.942	0.946	0.879	0.911	0.049

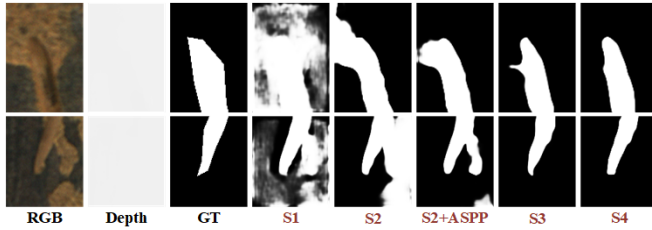


Fig. 7. The visual presentation of the ablation study on the main modules.

TABLE III
QUANTITATIVE COMPARISON WITH SEMI-SUPERVISED
SEMANTIC SEGMENTATION METHODS BASED ON
CONSISTENCY TRAINING [21]

Method	NEU RSDDS-AUG						
	Sm↑	W_Fm↑	meanEm↑	MaxEm↑	MeanFm↑	MaxFm↑	MAE↓
CCT _{Semi} ^{20%}	0.806	0.811	0.890	0.897	0.830	0.843	0.083
OURS_{Semi}^{20%}	0.846	0.871	0.931	0.934	0.879	0.892	0.058

4) *Ablation of Aux-Decoder Perturbation Settings:* Table Table V shows the quantitative metric comparisons under different perturbation mechanisms' hyperparameter settings used in our Aux-decoder (i.e., object mask perturbations and VAT perturbations). It can be observed that we employed perturbations more practical for this specific RGB-D Rail Surface Defect Inspection task.

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TABLE IV
QUANTITATIVE COMPARISON OF THE PROPOSED DSSNET UNDER
DIFFERENT PROPORTIONS OF LABELED SAMPLES

Settings	NEU RSDDS-AUG						
	Sm↑	W_Fm↑	meanEm↑	MaxEm↑	MeanFm↑	MaxFm↑	MAE↓
Semi ^{10%}	0.825	0.838	0.909	0.916	0.858	0.866	0.071
Semi ^{15%}	0.844	0.866	0.928	0.932	0.873	0.887	0.060
Semi^{20%}	0.846	0.871	0.931	0.934	0.879	0.892	0.058
Semi ^{25%}	0.852	0.874	0.933	0.935	0.884	0.894	0.056
Semi ^{30%}	0.854	0.875	0.935	0.937	0.885	0.897	0.055
Semi ^{40%}	0.858	0.881	0.937	0.939	0.889	0.901	0.053
Full^{100%}	0.866	0.892	0.942	0.946	0.897	0.911	0.049

TABLE V
EXPERIMENTS WITH DIFFERENT CONFIGURATIONS OF AUXILIARY
DECODER (AUX-DECODER). SPECIFICALLY, OM: OBJECT MASK
PERTURBATIONS, V: VAT PERTURBATIONS

OM-V	NEU RSDDS-AUG						
	Sm↑	W_Fm↑	meanEm↑	MaxEm↑	MeanFm↑	MaxFm↑	MAE↓
0-3	0.819	0.839	0.900	0.904	0.854	0.865	0.079
1-3	0.847	0.868	0.925	0.929	0.875	0.888	0.060
2-3	0.843	0.862	0.927	0.932	0.868	0.888	0.060
3-1	0.763	0.774	0.129	0.827	0.861	0.784	0.129
3-2	0.786	0.803	0.857	0.874	0.828	0.836	0.096
3-0	0.833	0.857	0.916	0.919	0.870	0.881	0.068
3-3	0.846	0.871	0.931	0.934	0.879	0.892	0.058

D. Generalization Validation

Our proposed DSSNet is designed for RGB-D rail surface defect detection. To further validate the generalization

TABLE VI

QUANTITATIVE COMPARISONS WERE CONDUCTED WITH 12 RELEVANT RGB-D SALIENT OBJECT DETECTION METHODS ON FIVE WIDELY USED DATASETS. WEAK: WEAKLY SUPERVISED RGB-D SOD METHODS, AND SEMI: SEMI-SUPERVISED RGB-D SOD METHODS

Methods	DUT				NLPR				SIP				STERE				NJUD			
	Em $\uparrow$	Fm $\uparrow$	W_F $\uparrow$	MAE $\downarrow$	Em $\uparrow$	Fm $\uparrow$	W_F $\uparrow$	MAE $\downarrow$	Em $\uparrow$	Fm $\uparrow$	W_F $\uparrow$	MAE $\downarrow$	Em $\uparrow$	Fm $\uparrow$	W_F $\uparrow$	MAE $\downarrow$	Em $\uparrow$	Fm $\uparrow$	W_F $\uparrow$	MAE $\downarrow$
MobileSal [40]	.936	.912	.869	.044	.953	.873	.875	.025	.908	.854	.827	.058	.919	.851	.860	.042	—	—	—	—
SPNet [63]	—	—	—	—	.954	.885	.891	.024	.932	.883	.869	.043	.929	.872	.880	.036	.935	.878	.903	.033
RD3D [68]	.949	.925	.913	.031	.957	.890	.894	.022	.919	.873	.852	.049	.927	.886	.871	.037	.918	.900	.890	.037
HAINet [58]	.937	.905	.887	.038	.951	.884	.884	.025	.924	.883	.860	.049	.930	.890	.871	.038	.916	.894	.881	.039
CDNet [69]	.937	.902	.879	.039	.951	.879	.885	.025	.914	.870	.840	.056	—	—	—	—	.903	.858	.829	.053
UCNet [65]	.934	.902	.878	.041	.948	.879	.875	.028	.922	.872	.852	.048	.922	.884	.873	.039	.866	.806	.767	.075
A2dele [70]	.924	.892	.870	.042	.945	.878	.867	.028	.893	.826	.794	.069	.914	.878	.856	.044	.897	.874	.851	.051
SSFNet [67]	.946	.915	.900	.033	.949	.875	.874	.026	.873	.781	.740	.088	.899	.840	.784	.065	.913	.886	.871	.043
JL-DCF [66]	.948	.917	.905	.032	.948	.869	.878	.024	.923	.881	.854	.050	.918	.868	.865	.040	.903	.858	.834	.056
JSM _{Weak} [41]	.870	.797	.698	.093	.888	.770	.692	.060	.781	.649	.546	.146	.852	.778	.688	.095	.788	.717	.599	.060
DENet _{Weak} [44]	—	—	—	—	.944	.851	.856	.032	.906	.804	.828	.063	.912	.846	.829	.051	.901	.856	.842	.051
OURS_{Semi}^{20%}	.956	.925	.915	.028	.953	.886	.882	.023	.910	.876	.846	.054	.933	.893	.883	.036	.919	.889	.874	.042

Methods	DUT				NLPR				SIP				STERE				NJUD			
	Sm $\uparrow$	maxE $\uparrow$	maxF $\uparrow$	MAE $\downarrow$	Sm $\uparrow$	maxE $\uparrow$	maxF $\uparrow$	MAE $\downarrow$	Sm $\uparrow$	maxE $\uparrow$	maxF $\uparrow$	MAE $\downarrow$	Sm $\uparrow$	maxE $\uparrow$	maxF $\uparrow$	MAE $\downarrow$	Sm $\uparrow$	maxE $\uparrow$	maxF $\uparrow$	MAE $\downarrow$
DSN _{Semi} ^{70%} [45]	.912	.927	.915	.045	.922	.942	.919	.032	.856	.886	.877	.070	.896	.911	.896	.041	.922	.927	.915	.045
OURS_{Semi}^{20%}	.925	.959	.934	.028	.912	.954	.902	.023	.868	.913	.881	.054	.902	.947	.905	.036	.893	.934	.895	.042

of our approach, we apply our method to address the task of RGB-D salient object detection. We utilize training data commonly used in existing methods [63], [65], [66], [67] and compare our approach with 12 recent RGB-D saliency detection methods on five widely used RGB-D SOD datasets. The comparison methods include nine fully supervised, two weakly supervised, and one semi-supervised method. During training, our DSSNet still employs 20% labeled samples and 80% unlabeled samples to ensure the alleviation of 80% annotation burden relative to fully supervised methods. As shown in Table VI, our DSSNet's generalization semi-supervised version achieves highly competitive results compared to the fully supervised methods. Meanwhile, when compared to semi-supervised methods [45] that alleviate 30% annotation burden, our performance in alleviating 80% annotation burden still exhibits a noticeable advantage. This thoroughly validates the effectiveness of our proposed DSSNet architecture and its high generalization.

V. CONCLUSION

In this paper, we present a depth-assisted semi-supervised network for RGB-D rail surface defect inspection. As the current fully supervised rail surface inspection methods based on deep learning suffer from two limitations: inadequate cross-modal information fusion and extremely time-consuming and labor-intensive demand for large amounts of labeled data. After sufficiently analyzing the above two limitations, we first build a semi-supervised learning architecture, which achieves a network that can guarantee no degradation in inspection performance with a small number of labeled samples and a high percentage of unlabeled samples supplied. Further, we propose a depth-assisted multi-scale cross-modal fusion module, which enables multi-scale mining and effective complementary fusion of cross-modal features. Eventually, the proposed DSSNet achieves highly competitive performance

compared to existing fully supervised methods and achieving over **97%** performance gains compared to our state-of-the-art fully supervised version while effectively alleviating **80%** of the annotation burden.

REFERENCES

- [1] M. Niu, K. Song, L. Huang, Q. Wang, Y. Yan, and Q. Meng, "Unsupervised saliency detection of rail surface defects using stereoscopic images," *IEEE Trans. Ind. Informat.*, vol. 17, no. 3, pp. 2271–2281, Mar. 2021.
- [2] W. Zhou, J. Hong, W. Yan, and Q. Jiang, "Modal evaluation network via knowledge distillation for no-service rail surface defect detection," *IEEE Trans. Circuits Syst. Video Technol.*, 2023.
- [3] L. Huang and A. Gong, "Surface defect detection for no-service rails with skeleton-aware accurate and fast network," *IEEE Trans. Ind. Informat.*, vol. 20, no. 3, pp. 4571–4581, Mar. 2024.
- [4] D. Zhang, K. Song, J. Xu, H. Dong, and Y. Yan, "An image-level weakly supervised segmentation method for no-service rail surface defect with size prior," *Mech. Syst. Signal Process.*, vol. 165, Feb. 2022, Art. no. 108334.
- [5] Q. Luo, X. Fang, L. Liu, C. Yang, and Y. Sun, "Automated visual defect detection for flat steel surface: A survey," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 3, pp. 626–644, Mar. 2020.
- [6] Q. Luo, Y. Sun, P. Li, O. Simpson, L. Tian, and Y. He, "Generalized completed local binary patterns for time-efficient steel surface defect classification," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 3, pp. 667–679, Mar. 2019.
- [7] H. Wang, J. Zhang, Y. Tian, H. Chen, H. Sun, and K. Liu, "A simple guidance template-based defect detection method for strip steel surfaces," *IEEE Trans. Ind. Informat.*, vol. 15, no. 5, pp. 2798–2809, May 2019.
- [8] Y. Wang, M. Niu, K. Song, P. Jiang, and Y. Yan, "Normal-knowledge-based pavement defect segmentation using relevance-aware and cross-reasoning mechanisms," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 4413–4427, Apr. 2023.
- [9] M. Nieniewski, "Morphological detection and extraction of rail surface defects," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 9, pp. 6870–6879, Sep. 2020.
- [10] G. Piao, J. Li, L. Udpa, J. Qian, and Y. Deng, "Finite-element study of motion-induced eddy current array method for high-speed rail defects detection," *IEEE Trans. Magn.*, vol. 57, no. 12, pp. 1–10, Dec. 2021.
- [11] D. Zhang, K. Song, J. Xu, Y. He, M. Niu, and Y. Yan, "MCNet: Multiple context information segmentation network of no-service rail surface defects," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.

- [12] X. Ni, H. Liu, Z. Ma, C. Wang, and J. Liu, "Detection for rail surface defects via partitioned edge feature," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 5806–5822, Jun. 2022.
- [13] Y. Wu, Y. Qin, Y. Qian, F. Guo, Z. Wang, and L. Jia, "Hybrid deep learning architecture for rail surface segmentation and surface defect detection," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 37, no. 2, pp. 227–244, Feb. 2022.
- [14] S. Ma, K. Song, M. Niu, H. Tian, and Y. Yan, "Cross-scale fusion and domain adversarial network for generalizable rail surface defect segmentation on unseen datasets," *J. Intell. Manuf.*, vol. 35, no. 1, pp. 367–386, 2022.
- [15] D. Zhang, K. Song, Q. Wang, Y. He, X. Wen, and Y. Yan, "Two deep learning networks for rail surface defect inspection of limited samples with line-level label," *IEEE Trans. Ind. Informat.*, vol. 17, no. 10, pp. 6731–6741, Oct. 2021.
- [16] J. Wang, K. Song, D. Zhang, M. Niu, and Y. Yan, "Collaborative learning attention network based on RGB image and depth image for surface defect inspection of no-service rail," *IEEE/ASME Trans. Mechatronics*, vol. 27, no. 6, pp. 4874–4884, May 2022.
- [17] J. Wu, W. Zhou, W. Qiu, and L. Yu, "Depth repeated-enhancement RGB network for rail surface defect inspection," *IEEE Signal Process. Lett.*, vol. 29, pp. 2053–2057, 2022.
- [18] J. Yang, W. Zhou, R. Wu, and M. Fang, "CSANet: Contour and semantic feature alignment fusion network for rail surface defect detection," *IEEE Signal Process. Lett.*, vol. 30, pp. 972–976, 2023.
- [19] B. Wang, W. Zhou, W. Yan, Q. Jiang, and R. Cong, "PENet-KD: Progressive enhancement network via knowledge distillation for rail surface defect detection," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.
- [20] Y. Yan et al., "Specificity autocorrelation integration network for surface defect detection of no-service rail," *Opt. Lasers Eng.*, vol. 172, Jan. 2024, Art. no. 107862.
- [21] Y. Ouali, C. Hudelot, and M. Tami, "Semi-supervised semantic segmentation with cross-consistency training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12674–12684.
- [22] J. Zhao, Y. Zhao, J. Li, and X. Chen, "Is depth really necessary for salient object detection?" in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 1745–1754.
- [23] P. Sun, W. Zhang, H. Wang, S. Li, and X. Li, "Deep RGB-D saliency detection with depth-sensitive attention and automatic multi-modal fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1407–1417.
- [24] Z. Zhang, J. Wang, and Y. Han, "Saliency prototype for RGB-D and RGB-T salient object detection," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 3696–3705.
- [25] S. Chen and Y. Fu, "Progressively guided alternate refinement network for RGB-D salient object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K.: Springer, 2020, pp. 520–538.
- [26] J. Wang, G. Li, J. Shi, and J. Xi, "Weighted guided optional fusion network for RGB-T salient object detection," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 20, no. 5, pp. 1–20, May 2024.
- [27] W. Zhou and J. Hong, "FHNet: Lightweight feature hierarchical exploration network for real-time rail surface defect inspection in RGB-D images," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–8, 2023.
- [28] H. Zhang et al., "MRSDI-CNN: Multi-model rail surface defect inspection system based on convolutional neural networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 11162–11177, Aug. 2022.
- [29] Y. Wu, M. Xing, Y. Zhang, Y. Xie, Z. Wu, and Y. Qu, "Perturbed progressive learning for semisupervised defect segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, 2023.
- [30] R. Ren, P. Shi, P. Jia, and X. Xu, "A semi-supervised learning approach for pixel-level pavement anomaly detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 9, pp. 10099–10107, Sep. 2023.
- [31] W. Zhou, Y. Zhu, J. Lei, J. Wan, and L. Yu, "CCAFNet: Crossflow and cross-scale adaptive fusion network for detecting salient objects in RGB-D images," *IEEE Trans. Multimedia*, vol. 24, pp. 2192–2204, 2021.
- [32] H. Wen, K. Song, L. Huang, H. Wang, and Y. Yan, "Cross-modality salient object detection network with universality and anti-interference," *Knowl.-Based Syst.*, vol. 264, Mar. 2023, Art. no. 110322.
- [33] J. Wang, K. Song, Y. Bao, L. Huang, and Y. Yan, "CGFNet: Cross-guided fusion network for RGB-T salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 5, pp. 2949–2961, May 2022.
- [34] L. Huang, K. Song, J. Wang, M. Niu, and Y. Yan, "Multi-graph fusion and learning for RGBT image saliency detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 3, pp. 1366–1377, Mar. 2021.
- [35] W. Zhou, Q. Guo, J. Lei, L. Yu, and J.-N. Hwang, "ECFFNet: Effective and consistent feature fusion network for RGB-T salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 3, pp. 1224–1235, Mar. 2022.
- [36] K. Song, J. Wang, Y. Bao, L. Huang, and Y. Yan, "A novel visible-depth-thermal image dataset of salient object detection for robotic visual perception," *IEEE/ASME Trans. Mechatronics*, vol. 28, no. 3, pp. 1558–1569, Jun. 2023.
- [37] J. Wang, K. Song, Y. Bao, Y. Yan, and Y. Han, "Unidirectional RGB-T salient object detection with intertwined driving of encoding and fusion," *Eng. Appl. Artif. Intell.*, vol. 114, Sep. 2022, Art. no. 105162.
- [38] K. Song, L. Huang, A. Gong, and Y. Yan, "Multiple graph affinity interactive network and a variable illumination dataset for RGBT image salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 7, pp. 3104–3118, Jul. 2023.
- [39] W. Zhou, Q. Guo, J. Lei, L. Yu, and J.-N. Hwang, "IRFR-Net: Interactive recursive feature-reshaping network for detecting salient objects in RGB-D images," *IEEE Trans. Neural Netw. Learn. Syst.*, 2021.
- [40] Y.-H. Wu, Y. Liu, J. Xu, J.-W. Bian, Y.-C. Gu, and M.-M. Cheng, "MobileSal: Extremely efficient RGB-D salient object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 10261–10269, Dec. 2022.
- [41] J. Li et al., "Joint semantic mining for weakly supervised RGB-D salient object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 11945–11959.
- [42] T. Yang, Y. Wang, L. Zhang, J. Qi, and H. Lu, "Depth-inspired label mining for unsupervised RGB-D salient object detection," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 5669–5677.
- [43] A. Li, Y. Mao, J. Zhang, and Y. Dai, "Mutual information regularization for weakly-supervised RGB-D salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 1, pp. 397–410, Jan. 2024.
- [44] Y. Xu, X. Yu, J. Zhang, L. Zhu, and D. Wang, "Weakly supervised RGB-D salient object detection with prediction consistency training and active scribble boosting," *IEEE Trans. Image Process.*, vol. 31, pp. 2148–2161, 2022.
- [45] X. Wang et al., "Boosting RGB-D saliency detection by leveraging unlabeled RGB images," *IEEE Trans. Image Process.*, vol. 31, pp. 1107–1119, 2022.
- [46] L. Zhu et al., "Self-supervised self-ensembling network for semi-supervised RGB-D salient object detection," *IEEE Trans. Multimedia*, vol. 25, pp. 676–689, Jan. 2023.
- [47] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11976–11986.
- [48] Z. Tu, Y. Ma, Z. Li, C. Li, J. Xu, and Y. Liu, "RGBT salient object detection: A large-scale dataset and benchmark," *IEEE Trans. Multimedia*, vol. 25, pp. 4163–4176, Oct. 2023.
- [49] A. Oliva and A. Torralba, "The role of context in object recognition," *Trends Cognit. Sci.*, vol. 11, no. 12, pp. 520–527, Dec. 2007.
- [50] T. Miyato, S.-I. Maeda, M. Koyama, and S. Ishii, "Virtual adversarial training: A regularization method for supervised and semi-supervised learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 8, pp. 1979–1993, Aug. 2018.
- [51] F. Milletari, N. Navab, and S. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
- [52] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, "Enhanced-alignment measure for binary foreground map evaluation," 2018, *arXiv:1805.10421*.
- [53] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4548–4557.
- [54] R. Margolin, L. Zelnik-Manor, and A. Tal, "How to evaluate foreground maps," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 248–255.
- [55] F. Perazzi, P. Krähenbühl, Y. Pritch, and A. Hornung, "Saliency filters: Contrast based filtering for salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 733–740.
- [56] R. Achanta, S. Hemami, F. Estrada, and S. Susstrunk, "Frequency-tuned salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1597–1604.
- [57] Y. Piao, W. Ji, J. Li, M. Zhang, and H. Lu, "Depth-induced multi-scale recurrent attention network for saliency detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 7254–7263.
- [58] G. Li, Z. Liu, M. Chen, Z. Bai, W. Lin, and H. Ling, "Hierarchical alternate interaction network for RGB-D salient object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 3528–3542, 2021.

- [59] N. Liu, N. Zhang, and J. Han, "Learning selective self-mutual attention for RGB-D saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13756–13765.
- [60] W. Ji, J. Li, M. Zhang, Y. Piao, and H. Lu, "Accurate RGB-D salient object detection via collaborative learning," in *Computer Vision–ECCV*. Glasgow, U.K.: Springer, 2020, pp. 52–69.
- [61] Y. Zhai et al., "Bifurcated backbone strategy for RGB-D salient object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 8727–8742, 2021.
- [62] M. Lee, C. Park, S. Cho, and S. Lee, "SPSN: Superpixel prototype sampling network for RGB-D salient object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Tel Aviv, Israel: Springer, 2022, pp. 630–647.
- [63] T. Zhou, H. Fu, G. Chen, Y. Zhou, D.-P. Fan, and L. Shao, "Specificity-preserving RGB-D saliency detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4681–4691.
- [64] F. Sun, P. Ren, B. Yin, F. Wang, and H. Li, "CATNet: A cascaded and aggregated transformer network for RGB-D salient object detection," *IEEE Trans. Multimedia*, vol. 26, pp. 2249–2262, Feb. 2024.
- [65] J. Zhang et al., "UC-Net: Uncertainty inspired RGB-D saliency detection via conditional variational autoencoders," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8582–8591.
- [66] K. Fu, D.-P. Fan, G.-P. Ji, and Q. Zhao, "JL-DCF: Joint learning and densely-cooperative fusion framework for RGB-D salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3052–3062.
- [67] M. Zhang, W. Ren, Y. Piao, Z. Rong, and H. Lu, "Select, supplement and focus for RGB-D saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3472–3481.
- [68] Q. Chen, Z. Liu, Y. Zhang, K. Fu, Q. Zhao, and H. Du, "RGB-D salient object detection via 3D convolutional neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 2, pp. 1063–1071.
- [69] W.-D. Jin, J. Xu, Q. Han, Y. Zhang, and M.-M. Cheng, "CDNet: Complementary depth network for RGB-D salient object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 3376–3390, 2021.
- [70] Y. Piao, Z. Rong, M. Zhang, W. Ren, and H. Lu, "A2dele: Adaptive and attentive depth distiller for efficient RGB-D salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 9060–9069.



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